NATIONAL ENGINEERING COLLEGE, K.R. NAGAR, KOVILPATTI – 628 503

Department of Computer Science and Engineering

Course: 23CS32E/23IT32E/23AD32E – UI/UX Design

Semester: Odd Semester 2025–2026

Experiential Learning Report

PROJECT TITLE: RESTAURANT MANAGEMENT SYSTEM

REG NO - NAME	2311123-AROCKYA ABINESH A 2311076-SRIDHAR H 2317021-VEERA PERUMAL S 2317041-SARAVANAN S
SEM/YEAR	Vth Sem/III Year
DEPARTMENT	CSE, IT, AI&DS, ECE
COURSE OUTCOME MAPPED	**CO5:** Design user-friendly, visually appealing interfaces and provide industry-relevant insights by applying UX design principles. *(PDL3)* **CO6:** Demonstrate creativity, diverse and inclusive attitude while practicing project component.

Faculty In-charge

TASK DESCRIPTION AND INITIAL PLAN:

The project aims to deliver a premium restaurant management application, focused on providing a seamless and visually engaging user experience throughout the entire holistic user journey, from registration to order tracking.

Problem Statement

Diners using typical restaurant applications often face a clunky, impersonal, and visually uninspired digital experience that fails to capture the quality and brand of the restaurant itself.

- **The Fragmented User Experience:** Customers frequently encounter disjointed navigation and complicated, multi-page checkout processes, creating friction that can lead to cart abandonment.
- **The Engagement Deficit**: Many applications lack the dynamic feedback and satisfying micro-interactions that make a digital experience feel modern and responsive. Key actions often feel static and fail to provide the user with a sense of confirmation or delight.
- **Poor Visual Presentation**: Inconsistent branding, a lack of clear visual hierarchy, and unappealing design can make the menu seem less attractive and the overall brand feel unprofessional, failing to showcase high-quality food photography effectively.
- **Post-Order Uncertainty:** After placing an order, users are often left without clear, real-time feedback on its status, leading to a disconnected and anxious waiting period.

Objective

To develop a premium restaurant management application that:

- Streamlines the entire customer journey, from initial browsing and ordering to a simplified, single-screen checkout and live order tracking.
- Creates a highly engaging and intuitive user interface through fluid animations, immediate visual feedback, and delightful micro-interactions.
- Establishes a strong and memorable brand identity with a polished, pixel-perfect visual design and a consistent, component-based system.

.

Phase	**Focus and Key Features**
Phase 1: Foundation & Core UI	**User Flow:** Implement the essential user journey, including the onboarding splash screen, login/registration, and menu browsing . UI Foundation: Establish the high-contrast dark theme, visual hierarchy with card components, and the system of reusable buttons and input fields . **Basic Interaction:** Integrate foundational UI animations like button state changes on press and elegant loading indicators.

Phase 2: Enhanced Ordering & Engagement	**Streamlined Checkout:** Implement the refined, single-screen checkout process with animated toggles for options like "Dine In" and "Take Away" . **Advanced Micro-Interactions:** Integrate delightful feedback moments, such as the "fly to cart" animation and the celebratory confetti burst on the order confirmation screen . **Brand Polish:** Implement the full set of custom, wireframe-style icons to ensure brand consistency across all screens.
Phase 3: Post-Order Experience & Scalability	**Live Order Management:** Develop the live order tracking screen featuring animated countdown timer to manage user expectations effectively . **Interactive Home:** Implement dynamic home screen elements, including the horizontally sliding promotional banner carousel . **System Handoff:** Finalize the component-based design system to ensure the design is scalable and ready for an efficient handoff to the development team.

1. <u>**USER RESEARCH & PERSONA:**</u>

- **Interview Summary:** Our research confirmed that while basic ordering functionality is expected, the primary differentiators for users are the **quality of the digital experience** and the **clarity of the process**. A seamless, visually engaging interface is not just a bonus; it's a key factor in building brand trust and encouraging repeat orders.

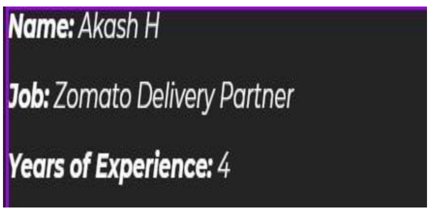
- **Visuals Drive Decisions:** Users strongly associate a polished, professional app with a high-quality restaurant. A modern, high-contrast dark theme that makes food photography look appealing was identified as a significant factor in their decision to order from a menu.

- **Frictionless Checkout is Non-Negotiable:** The most significant frustration highlighted by users is a long or confusing checkout process. A streamlined, **single-screen system** where they can review their order and pay without navigating multiple pages is considered an essential feature to prevent them from abandoning their cart.

- **Desire for an Engaging, Responsive Experience:** Users expressed a clear preference for an interface that feels alive and provides immediate, satisfying feedback. Features like **celebratory micro-interactions** on order confirmation or seeing an item visually move to the cart were described as key elements that make the experience feel more delightful and less transactional.

- **Clarity Over Post-Order Anxiety:** The waiting period after payment is a major source of user anxiety. Providing **live order tracking** with a real-time countdown timer was universally seen as a critical feature for managing expectations and improving the overall service experience.

- **User Personas:**



Basic Utility: The app's core idea of managing restaurant information (pickup locations, wait times, menu status) is fundamentally useful for a delivery partner.

Cons & Suggestions:

- **Lack of Emergency Features:** This points out a major gap. The app needs features to locate the nearest **hospitals**, **police stations**, or **roadside assistance** in case of an accident or breakdown, a key concern for a driver on the road.
- **Needs Essential Service Locators:** Beyond restaurants, it highlights the difficulty in finding **public restrooms**, **petrol pumps**, or **bike repair shops**. Adding these locators would be a valuable addition.
- **Limited Scope:** The feedback implies that for a delivery partner, an app focused *solely* on restaurant data is too basic and misses other critical real-world needs (like safety and essential services).

Name: *Vishal M*

Job: *Swiggy Delivery Partner*

Years Of Experience: *5*

Basic Utility: He acknowledges that the app's core idea of receiving orders, navigating to restaurants, and tracking customer locations is fundamentally useful for his job.

Cons & Suggestions:

- **Lack of Emergency Features:** He points out a major gap. The app needs a one-tap "SOS" button to contact Swiggy support or local authorities in case of an accident or security threat, a key concern when working late nights.
- **Needs Essential Service Locators:** Beyond restaurants, he highlights the difficulty in finding designated **waiting zones** (especially during rain) or **partner-friendly eateries** for his own breaks, and suggests this would be a valuable addition.
- **Limited Scope:** His feedback implies that for a delivery partner, an app focused solely on active orders is too basic. It misses

Name: *SD Ponseelan*

Job: *Owner Of SDR Restuarant*

Years Of Experience: *20*

Basic Utility: He acknowledges that the dashboard's core idea of receiving and confirming online orders is fundamentally useful for managing his business.

Cons & Suggestions:

- **Lack of Emergency Features:** He points out a major gap. The app needs an "Emergency Stop" button to instantly pause all new orders in case of a kitchen fire, equipment failure, or sudden staff shortage, a key concern during peak dinner service.
- **Needs Essential Service Locators:** Beyond just accepting orders, he highlights the difficulty in **contacting the assigned delivery partner directly** (when an order is complex) or **reporting an issue with a specific partner**, and suggests this would be a valuable addition.
- **Limited Scope:** His feedback implies that for a restaurant owner, a dashboard focused solely on *new* orders is too basic. It misses critical real-world needs like **managing inventory** (to mark items "out of stock" in real-time) and **viewing delivery partner ETAs** (to ensure food is packed hot and fresh).

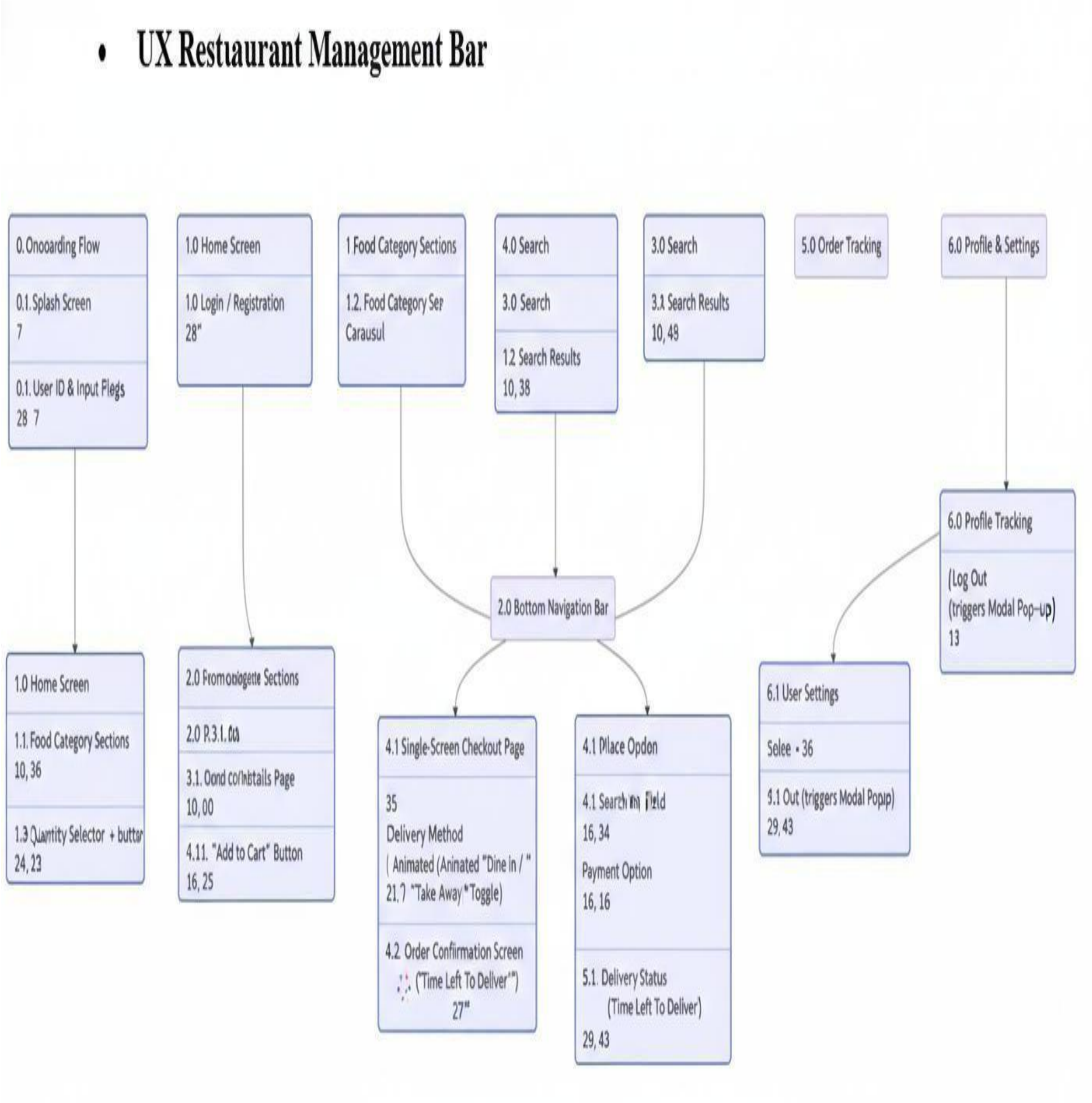
2. USER JOURNEY MAPPING & UX SITE MAP:

User Journey Map & Description

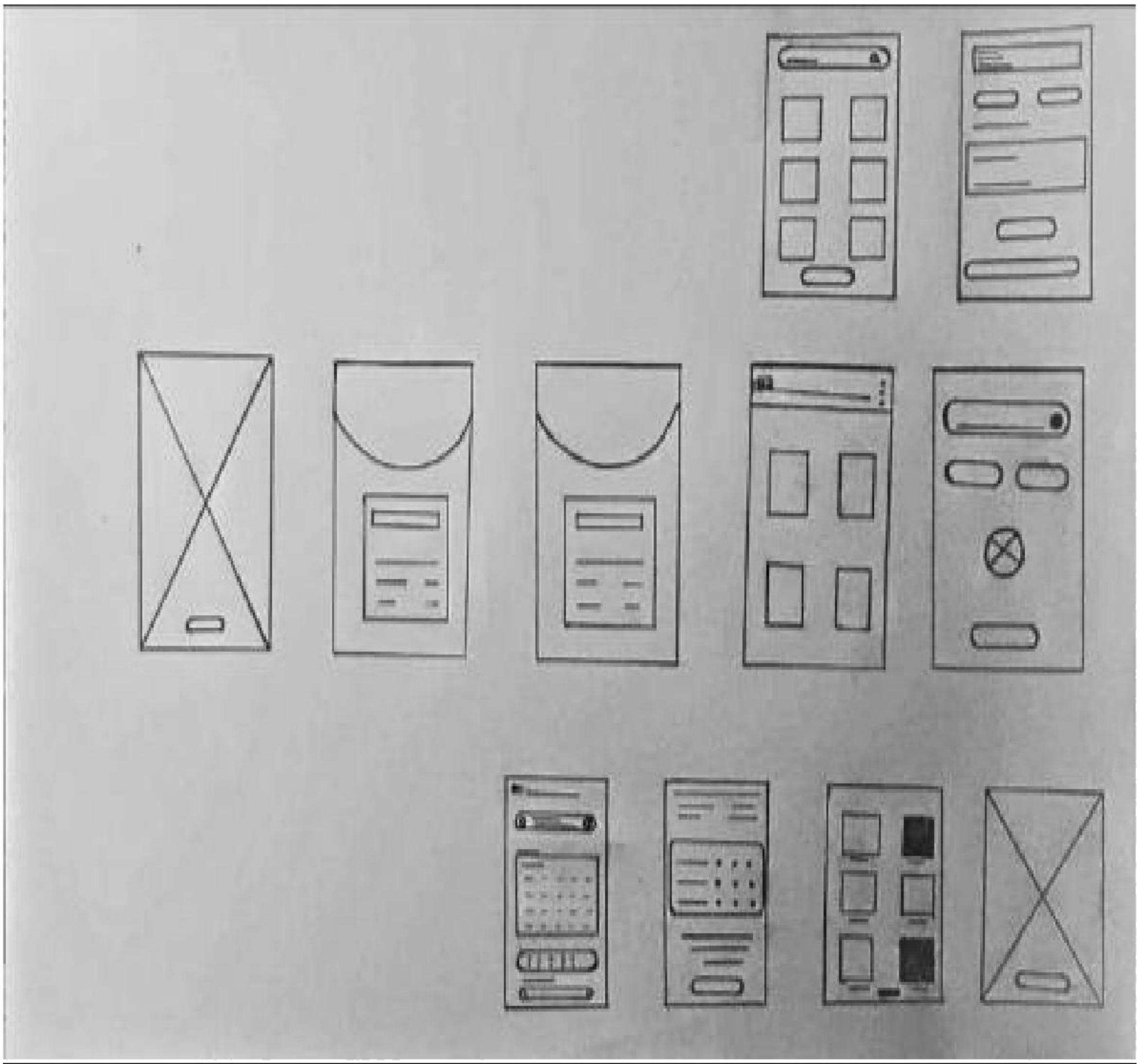
The map focuses on transforming the common points of friction in a digital ordering process into a seamless and delightful experience.

Journey Stage	Past / Present Pain Point	Future Experience (With "Bites Up" App)
1. Browsing & Discovery	**An unappealing or inconsistent visual design makes the food and brand seem unprofessional.**	**The user experiences a premium, modern aesthetic with a high-contrast dark theme that enhances high-quality food photography[1].**
2. Ordering & Checkout	**A long, multi-page checkout process creates friction and leads to users abandoning their carts[2].**	**The checkout is a streamlined, single, clear screen [3], and adding items is confirmed with a satisfying animation of the item flying to the cart icon[4].**
3. Post-Order & Waiting	**Uncertainty and anxiety after placing an order due to a lack of status updates.**	**The user gets clear, real-time feedback from a live order tracking screen featuring an animated countdown timer[5].**
4. Order Confirmation	**A simple, transactional confirmation screen that is unmemorable and anticlimactic.**	**The confirmation is transformed into a "moment of delight" [6]with a checkmark that appears in a "celebratory confetti burst" for positive reinforcement[7].**

3. UX Site Map Diagram:



4. LOW-FIDELITY WIREFRAME, GRID & LAYOUT PLANNING:



Grid system used:

- The design clearly divides the screen space to display content blocks. The main dashboard **(Frame 4)** uses a 2x2 grid for primary navigation (e.g., "Manage Orders," "Menu Items," "Reservations," "Analytics") in a consistent and modular fashion.
- On the content pages (e.g., "Menu" or "Staff List"), the use of a multi-column card layout **(Frame 10)** and stacked list-item layout **(Frame 6)** suggests a flexible grid structure that efficiently maximizes screen real estate while ensuring scannability and visual balance.

Authentication & Onboarding

- **Sequential Flow:** The app establishes a clear entry path, starting from a splash screen **(Frame 1)** and moving directly to unambiguous Login **(Frame 2)** or Sign-Up **(Frame 3)** screens, ensuring immediate access for registered managers.

Information Display

- **Task-Based Structure:** Detailed information is segmented into scannable modules. For example, the "Reservations" module uses a card-based structure with a dedicated calendar **(Frame 8)** and list view **(Frame 11)**, resolving the issue of managing bookings efficiently.
- **Data-Entry Focus:** Screens requiring input, like "Edit Menu Item" or "Order Details" **(Frame 7)**, use a simple, single-column layout with stacked form fields. This minimizes clutter and guides the manager through the data entry process logically.
- **Clear Call-to-Action:** Nearly every functional screen **(Frames 2, 3, 5, 8, 10)** features a prominent, full-width button at the bottom. This provides a clear primary action (e.g., "Login," "Confirm Booking," "Save Item"), which is critical for a fast-paced restaurant environment.

5. HIGH-FIDELITY WIREFRAME & UI KIT:

Components in UI Kit used:

1. **Color Palette** The primary palette is functional and high-contrast, designed for quick scannability and clear actions in a fast-paced restaurant environment.

Component	Observed Color	Context / Use
Primary Background	Pure White / Off-White	Used for the main screen background to maximize contrast and readability of order details and data.
Container / Card Background	Light Gray	Used for individual order cards, menu item containers, and dashboard modules to visually separate them from the main background.
Primary Brand / Accent	Professional Blue	Used for key interactive elements like the active navigation tab, primary buttons ("Confirm Order," "Add Item"), and selected states.
Semantic Success	Bright Green	Used specifically for status indicators ("Online," "Paid," "Delivered") and confirmation buttons to provide immediate positive feedback.
Semantic Alert	Bright Red / Orange	Used for critical warnings, error messages, and negative actions like "Cancel Order," "Offline," or "Out of Stock" toggles.
Primary Text / Icons	Dark Gray / Black	Used for all body text, headings, order details, and most primary icons to ensure maximum legibility.

2. Typography (Fonts)

The font choices support high legibility and a clean, functional tone, consistent with the project goal of enabling fast and efficient operations.

- **Primary Headings:** A clear, bold **sans-serif** font (e.g., similar to **Inter Bold** or **Roboto Condensed**) is used for major headings like the "Dashboard" title, "Live Orders," and "Menu Management." This lends a modern, professional feel and allows staff to quickly scan and identify different sections.
- **Body/UI Text:** A clean, highly legible **sans-serif** font (e.g., **Roboto Regular** or **SF Pro Text**) is used for all smaller text, order details, table data, and interface labels (e.g., "Customer Name," "Mark as Ready," "Add New Item") to ensure high readability and reduce errors in a fast-paced, data-rich environment.

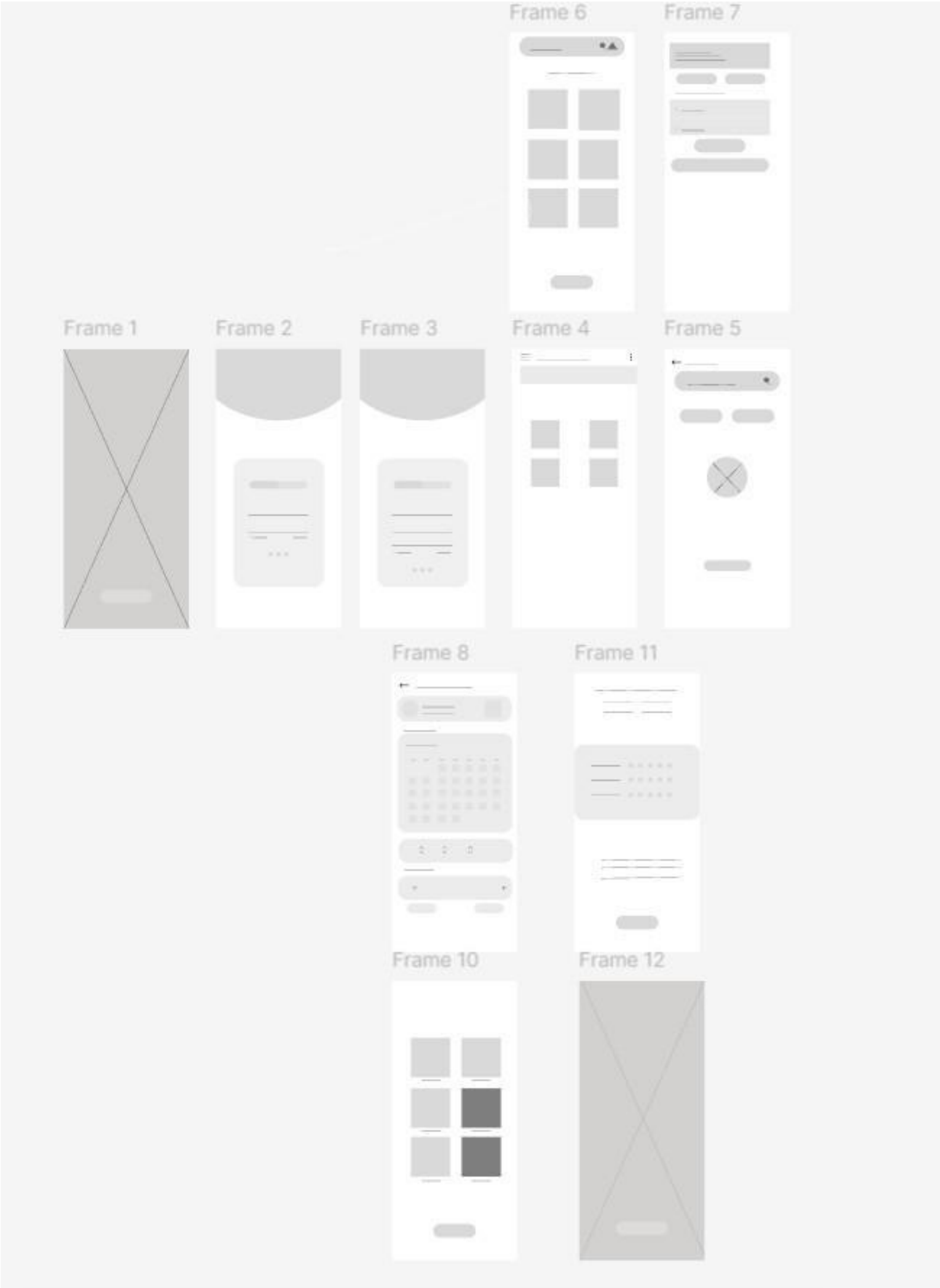
3. Buttons & Interactive Elements

The design uses distinct styles for primary actions, toggles, and navigation, prioritizing speed and clarity.

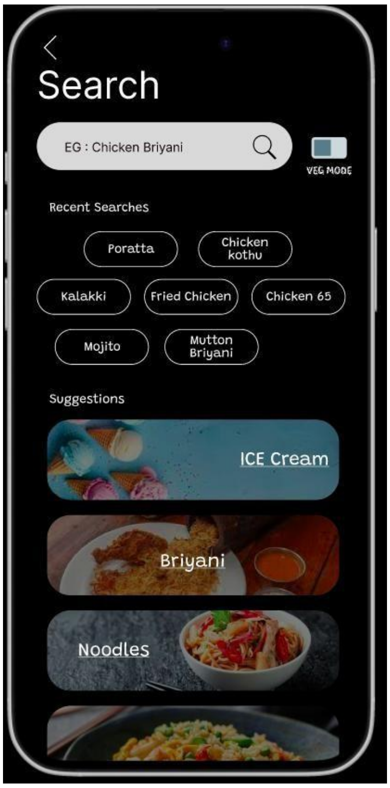
- **Call-to-Action (CTA) Links:** Links like "Edit Menu Item" or "View Order Details" often use a clean text style in the primary blue accent color, sometimes with a simple icon (like a pencil or eye) to indicate the action.
- **Card Buttons / Toggles:** The "In Stock / Out of Stock" control is integrated directly into each menu item card. It functions as a clear **switch toggle**, with its color change (e.g., Green for "In Stock," Gray for "Out of Stock") providing immediate visual feedback.
- **Primary Navigation (Sidebar):** The main restaurant menu is accessed via a fixed vertical sidebar. It uses a clean list of icons and text labels (e.g., "Live Orders," "Menu," "Reservations"), with the active screen highlighted by a solid color bar or background.
- **Primary Action Buttons:** Main buttons like "Accept Order" or "Save Changes" are visually striking: large, rectangular buttons with a solid fill using the **success (green)** or **primary (blue)** color, clearly guiding the user to the most important action.
- **Status Toggles:** The main "Restaurant Status" control (e.g., "Online / Offline") is a large, prominent switch toggle in the app's header, using a strong color change (green/red) to make the restaurant's live status unmistakable at a glance.

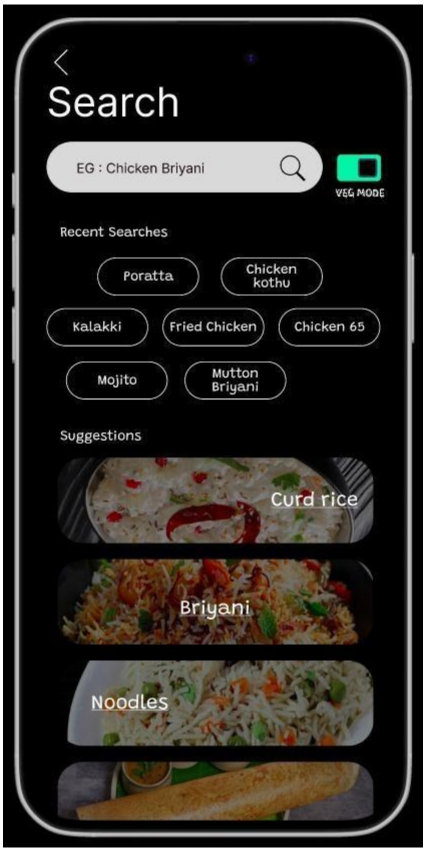
Order Card Actions: Uses small, distinct buttons (e.g., "Mark as Ready," "Assign Driver") integrated directly at the bottom of each order card, keeping controls contextual and minimizing clicks.

6. HI-FI Wireframes:



7. INTERACTIVE PROTOTYPE:




Search
EG : Chicken Briyani
VEG MODE
Recent Searches
Poratta
Chicken kothu
Kalakki
Fried Chicken
Chicken 65
Mojito
Mutton Briyani
Suggestions
Curd rice
Briyani
Noodles


Your Cart
Pepper Chicken Gravy
Quantity
- 1 +
₹99
Kothu Poratta
Quantity
- 2 +
₹149
Place Order
More Like This:
Chicken Shawarma
- 3 +
Butter Chicken Naan
- 3 +
Pepper Chicken Fry
- 3 +


Dine In
Delivery
Your Token No:
49
Previous Token
46
Next Token
47
Your Orders
Bun Porrata (chicken Gravy)
chicken 65 (Boneless)
Coke (3Combo)
Ice Cream Sandwitch
"First we eat, then we do everything else." - M.F.K. Fisher
Yum Run


Dinner Bell
Updates
Delivery
The Delivery boy has been reached your location
make sure to check out ... !
12:34 PM
Delivery
Your order has been taken by
Name:Sridhar H
Ph.no:123456789
Your has been left the restaurant
12:25 PM
Delivery
Your order has been Placed ID:1452kn6
Payment: Cash On Delivery
Waiting for the delivery to claim it
12:15 PM
Dine In
Your Order is ready SIr/Madam
Be Ready May you be the next Token Number
8:47 PM
Dine In
Payment Successful for Your Order
1234bc4
Check your Token number in "Yum run" menu
8:38 PM
Dine In
Your Order 1234bc4 has been Placed ... !
kindly make your payment through UPI
8:34 PM
Dinner Bell

- Tool Used: Figma

8. UI ANIMATIONS:

Description of animations:

- **Description of animations:** The prototype uses motion to create clarity, guide attention, and provide critical feedback.

- **New Order Animation:** The new order cards on the "Live Orders" screen **pulse or flash** with a green accent when they first appear. This is a high-priority animation used to immediately draw staff attention to a new task.

- **Page Transition and Status Change:** Content sections (like moving an order from "New" to "In Progress") **slide horizontally** across columns. This reinforces the workflow and provides a clear visual confirmation of the status change.

- **Click/Tap Feedback (Intention):** The design specifies that when the user presses a primary button (like "Accept Order" or "Mark as Ready"), the button should **subtly shrink or darken** to signal the system has registered the tap before the next action occurs.

Demo links:
https://www.figma.com/design/xVOinEPcZfDNvnq9sKFXAS/Food-Delivery-App?node-id=0-1&t=TNqzvXFycNJUB0s6-1

9. MICRO-INTERACTIONS & ITERATIVE TESTING:

a. **Interactions Implemented:** The following micro-interactions provide clear, instantaneous feedback to user actions, improving reliability and operational speed.
b. **Accept Order Interaction:** When the user clicks the "Accept" button (Action), the button itself shows a brief loading state, and the entire order card instantly animates, sliding from the "New Orders" column to the "In Progress" column (Feedback), confirming the action has succeeded.
c. **Out of Stock Toggle:** When the user clicks the toggle on a menu item (Action), the switch itself animates by sliding from one side to the other. Crucially, the entire menu item card instantly grays out and a small notification appears saying "Item marked unavailable" (Feedback).
d. **Order Details Expansion:** Clicking the small "View Details" arrow on a compact order card (Action) animates the card, expanding it vertically to display customer notes, full address, and item modifications (Feedback).
e. **Restaurant Status Toggle:** Clicking the main "Online/Offline" switch provides feedback by sliding the toggle and instantly changing the color of the entire app header (e.g., from green to red), signaling the restaurant's live status.

10. FINAL DESIGN REFINEMENT:

- Final UI Snapshots:

    - **Home Screen:** Features a clean, neutral background contrasting with a large,
    - **Live Orders Dashboard (Home Screen):** Features a high-contrast, clean interface. The screen is divided into vertical columns (e.g., "New Orders," "In Progress," "Ready for Pickup"). New order cards use a **pulsing green animation** to draw immediate attention. Each card has clear, bold typography and contextual buttons ("Accept," "Mark as Ready").
    - **Menu Management Page:** This page features a clean list or grid of all menu items. Each item has a title, price, and a prominent **"In/Out of Stock" toggle switch**. The layout uses white space and light gray card backgrounds to clearly separate items, ensuring high scannability for quick updates.
    - **Restaurant Settings Page:** The layout is functional and clean, presenting operational settings in a vertical, list-like format. Options like "Manage Staff," "Edit Store Hours," and "Notification Settings" are clearly labeled, often with icons. The master **"Restaurant Online/Offline" toggle** is given top priority, using a large size and strong color change (green/red).
    - **Analytics & Reports Page:** Uses a clean dashboard layout with data visualizations. It features sections for "Top Selling Items," "Peak Order Times," and "Revenue Summary," presented as simple bar charts and line graphs for at-a-glance insights.
    - **Order Detail Page:** Displays all information for a single order: customer details, item list (with modifications), and payment status. It often includes an integrated **small map view** showing the customer's address and, if applicable, the assigned driver's real-time location.

    - The project is a **Restaurant Management System**, developed as a high-fidelity UI/UX prototype. It is conceived as a high-efficiency operational tool designed to streamline all front-of-house digital operations for busy restaurants.
    - **Core Function:** The system addresses the critical issue of slow, complex, or error-prone order handling. It provides a single, unified interface for managing incoming orders, updating menu availability, and tracking order status from acceptance to handoff.
    - **Key Features:** Includes a real-time **Live Order Dashboard** with status columns, integrated **Menu & Inventory Management** (featuring one-tap "In/Out of Stock" toggles), and clear, actionable notifications for staff.
    - **Design Philosophy:** The final prototype utilizes Figma to showcase a task-oriented and user-centric design. It features functional animations (like order cards **sliding** between status columns) and strong micro-interactions (e.g., the "Item marked unavailable" confirmation) to enhance staff efficiency, reduce cognitive load, and minimize operational errors.

OUTCOMES & LEARNINGS

Skills Gained

The experiential project provided hands-on experience in:

- **User-Centred Design Principles:** Mapping user needs (like safety and efficiency) to tangible features.
- **Prototyping:** Developing a high-fidelity, interactive prototype using Figma.
- **Micro-interaction Design:** Applying small animations and feedback mechanisms to create a delightful and intuitive user experience.
- **Information Architecture:** Structuring verified, dense travel information for improved readability and flow.

Tool Used

The primary tool used for design and interactive prototyping was Figma.

Real-World Relevance

The Restaurant Management System project provides a highly relevant solution to critical modern restaurant operation problems. It specifically addresses:

- The Operational Bottleneck: By focusing on a real-time, consolidated dashboard, it solves the primary pain point of missed orders, kitchen confusion, and errors during peak service hours.
- Inventory and Accuracy: The emphasis on one-tap "In/Out of Stock" toggles builds customer trust and reduces order failures. It directly prevents the negative customer experience of ordering an unavailable item.
- Business Insight: The design anticipates and solves the manager's need for simple, actionable data (like "Top Selling Items" and "Peak Hours"), shifting the app from a simple order-taker to a core business intelligence tool.